

Fabrication of amphiphilic self-healing coatings with high mechanical properties and their antifouling performance

Ziyuan Liu, Chunju He^{*}

The State Key Laboratory for Modification of Chemical Fibers and Polymer Materials, College of Materials Science and Engineering, Donghua University, Shanghai 201620, PR China

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ABSTRACT

A self-healing antifouling coating with excellent mechanical strength was prepared via a simple “one-step” method. A series of well-designed amphiphilic block copolymer polydimethylsiloxane-poly(2-(dimethylamino) ethyl methacrylate) (PDMS-PDMAEMA) synthesized by atom transfer radical polymerization was introduced to the self-healing polydimethylsiloxane-based polyurethane (PDMS-PU) system, which was further zwitterionized during solidification. The slight swelling resulting from zwitterionic segments promoted the contact fusion rate of the mechanically damaged parts, which improved the underwater self-healing ability of the resulting coating. The PDMS segments in both amphiphilic polymer and PU were entangled with each other to form a semi-interpenetrating network while the zwitterionic segments migrated externally when being immersed underwater, thus a micro-phase separation structure was obtained for antifouling coating. Qualitative and quantitative tests have also proved that the coating can significantly resist the adhesion of proteins ($156 \mu\text{g}/\text{cm}^2$ vs $14 \mu\text{g}/\text{cm}^2$) and marine microorganisms ($425 \times 10^3 \text{ n}/\text{cm}^2$ vs $12 \times 10^3 \text{ n}/\text{cm}^2$). In addition, the organically modified montmorillonite (MMT) effectively improved the coating's mechanical properties via coordination bonding, as the resulting coating's fracture strength was increased by 50 % compared with the initial PU coating. In this study, an inherent problem in underwater self-healing antifouling coatings is solved perfectly, that is to achieve self-healing properties without loss of mechanical properties, which provides inspiration for the research and development of novel self-healing antifouling coatings.

1. Introduction

Based on the requirements of environmental protection and durable antifouling effectiveness, there has been considerable interest in multi-functional coatings to inhibit marine bio-fouling [1–3]. Low surface energy coatings i.e. poly(dimethylsiloxane) (PDMS) have always been a research hotspot, since they have high elasticity and excellent underwater stability, and no substance is released to the marine environment and the formed fouling is easily removed by shearing force due to its weak surface adhesion [4–6]. Unfortunately, PDMS-based coatings exhibit poor antifouling performance under static conditions [7–9]. In addition, PDMS has poor mechanical properties and adhesion to the substrate, which is easily damaged and difficult to be repaired once damaged, thus limiting its further applications [10–12]. Therefore, it is meaningful to develop a new type of PDMS-based antifouling material with excellent static antifouling ability and self-healing capability.

A series of self-healing PDMS-based materials have been prepared.

He et al. [13] incorporated zwitterionic polymers modified gallium-based liquid metal (GLM) to PDMS coating, whose antibacterial and surface hydration property synergistically endowed the coating with significant antifouling properties. Meanwhile, GLM promoted the cross-linking of unreacted vinyl groups in PDMS, bringing self-healing properties to the coatings. Yu et al. [14] prepared a polysiloxane-polyurea elastomer antifouling coating with lubricant self-replenishment properties, where the silicone oil encapsulated could be released controllably under mechanical stimulation.

In general, self-healing ability and mechanical strength are mutually exclusive in nature. How to improve mechanical properties while maintaining its self-healing efficiency is a huge challenge for self-healing coatings [15–17], while the addition of various inorganic fillers has opened a promising avenue. Liu et al. [18] introduced acrylic-modified bentonite to a cross-linking network via its copolymerization with acrylic acid. The tensile strength of the resulting organic-inorganic hydrogel was greatly improved. Qi et al. [19] incorporated a 2D

^{*} Corresponding author.

E-mail address: chunjuhe@dhu.edu.cn (C. He).

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lamellar montmorillonite in a self-healing polyvinyl alcohol-tannic polymer coating and the resulting coating showed excellent mechanical properties.

Considering the complexity of actual marine environment, the requirements for underwater coating materials are particularly tough. Since coordination bonds or hydrogen bonds are sensitive to water molecules, resulting in the failure of self-healing function, it is difficult to prepare a marine antifouling coating with both high self-healing efficiency and high mechanical strength. Shang et al. [20] deposited $\text{SiO}_2/\text{PDA}/\text{Ag}$ particles on the surface of acrylic resin/poly(ethylene-vinyl acetate) coating. The resulting hybrid coating with underwater super-oleophobicity exhibited excellent healing properties and antifouling capabilities while resisting oily liquids and bio-organism fouling. Liu et al. [21] constructed a novel graphene-epoxy nanocomposite self-healing coating, which can be easily healed when suffering damage by the dynamic host-guest recognition between cyclodextrin and adamantane. Though the above self-healing coatings have antifouling ability, their mechanical properties have not been characterized, and it is difficult to predict their durability underwater.

As we all know, zwitterions, i.e. sulfobetaine, carboxy betaine, etc. have excellent water-binding ability to form a dense hydration layer on substrate surface underwater [22–24], so as to greatly decrease the attachment density of fouling organisms. However, the pure zwitterionic polymer coatings have poor mechanical properties and durability underwater [25]. It is also difficult for zwitterions to incorporate with low surface energy materials due to their huge difference in polarity, which limits their further applications.

In this study, the zwitterions were introduced in PDMS-PU to endow the resulting self-healing coating with antifouling performance, and the incorporation of organically modified montmorillonite further improved the mechanical properties. The design strikes a balance among mechanical properties, self-healing efficiency, and antifouling functionality of the coating, thus perfectly solving the contradiction between mechanical performance and self-healing efficiency of underwater antifouling coating, which may be a pioneer in the field of underwater self-healing composite antifouling coatings.

2. Experimental section

2.1. Materials

Br-PDMS-Br macroinitiator was synthesized according to our previous work [26]. 2-(Dimethylamino) ethyl methacrylate (DMAEMA) from Shanghai Yishi Chemical Co., Ltd. was purified using an alkaline silica column to remove the inhibitor. Copper(I) bromide was purchased from Shanghai Chemical Reagent Plant and was purified with acetic acid according to a standard procedure. Isopropyl alcohol and N, N, N', N''-pentamethyldiethylenetriamine (PMEDTA) was used directly without further purification.

Aminopropyl terminated poly(dimethylsiloxane) (PDMS-NH₂) (M_w = 4000 g/mol) from Shanghai Yishi Chemical Co., Ltd. was dried under reduced pressure for 2 h before use. Tetrahydrofuran (THF, 99 %) from Aladdin was refluxed over CaH₂ and distilled before use. Isophorone diisocyanate (IPDI, 99 %), trimethylolpropane (TMP), cetyl trimethyl ammonium bromide (CTAB), (CH₃COO)₂Cu and 1,3-propane sultone from Aladdin was used as received. Montmorillonite (MMT) was purchased from Shandong Yousuo Chemical Technology Co., Ltd. and was used as received.

Bovine serum albumin (BSA) and lysozyme were purchased from Shanghai Chemical Reagent Company (Shanghai, China) and used as received. *Phaeodactylum tricornutum* was supplied by the College of Marine Science and Engineering, Qingdao Agricultural University. Artificial seawater (ASW) was prepared according to ASTM D1141. Commercial PDMS (sylgard 184) coating is purchased from Dow Corning (China) Co., Ltd.

2.2. Synthesis of the copolymer

In this section, a series of PDMS-PDMAEMA triblock polymers were synthesized by atom transfer radical polymerization (ATRP), and their molecular weight and block ratio were adjusted via changing the feed ratio of initiator to initial monomer.

The typical synthesis procedure for the triblock polymer was as follows, a solution of a certain amount of DMAEMA, Br-PDMS-Br macroinitiator (1 g, 0.25 mmol), PMEDTA (0.0865 g, 0.5 mmol) and isopropyl alcohol (16 mL) was sealed in a double-neck flask equipped with a magnetic stirring bar, degassed by three freeze–pump–thaw cycles. Copper(I) bromide (0.036 g, 0.25 mmol) was then added under nitrogen atmosphere, the solution was placed in 60 °C oil for another 24 h. The obtained product was precipitated 3 times in deionized water and dried at 60 °C for 12 h. The resulting solid polymers were named **DPD_x**, where x presented DMAEMA content calculated according to the NMR and GPC result (Fig. S1 and Table S1), and the product was characterized by FTIR as shown in Fig. S2.

2.3. Organic modification of MMT

25 g MMT powder was dispersed in HCl solution (0.01 M) at 60 °C for 3 h, the product was washed with deionized water and drying and named H-MMT. A 5 mg/mL H-MMT suspension was obtained by dispersing 5 g H-MMT into 1000 mL deionized water under vigorous stirring for a week. 20 g CTAB solution (0.3 M) was added to the above suspension under mechanical stirring at 80 °C for 24 h. CTAB-MMT was obtained by suction filtered and washed with deionized water until no chloride ion was detected (AgNO₃ solution test). A certain amount of CTAB-MMT reacted with (CH₃COO)₂Cu solution (0.15 wt%) according to the same procedure as above, the resulting product was named Cu-MMT and dispersed in THF at a concentration of 3 wt% for following exploration.

2.4. Preparation of coatings

PDMS-NH₂ (8.0 g, 2 mmol) and DPD_x (0.01 mmol, the masses are 0.15 g, 0.25 g, 0.34 g, 0.56 g, respectively according to different x) were dissolved in 14 mL Cu-MMT suspension, IPDI (1.167 g, 5.25 mmol) and a certain amount of 1,3-propane sultone solution (0.1 M) was slowly added to the above solution under nitrogen atmosphere. The mixture was kept stirring at room temperature for 12 h before the addition of TMP (0.067 g, 0.5 mmol), and the mixture was heated to 85 °C for 4 h, casted on substrate or poured in a polytetrafluoroethylene mold, then dried at 90 °C for 48 h. The resulting samples were named **(Cu)MMT-S_xPU** (x is 34, 66, 96 and 165 respectively corresponding to DPD_x). As controls, samples without MMT and 1,3-propane sultone were named **D_xPU**, samples without MMT were named **S_xPU**. In addition, samples prepared from PDMS-NH₂, IPDI and TMP only was named **PU**.

3. Characterization

3.1. Proton nuclear magnetic resonance spectroscopy (¹H NMR)

The ¹H NMR spectrum was recorded on an AV 600 MHz spectrometer (Bruker) using CDCl₃ as solvent and tetramethylsilane (TMS) as internal standard.

3.2. Gel permeation chromatography (GPC)

GPC was performed at room temperature on a BI-MwA Gel Permeation Chromatography (Waters, Milford, MA) equipped with a light scattering instrument (Brookhaven, Holtsville, NY), using THF as the eluent at a flow rate of 0.8 mL/min with a polystyrene standard.

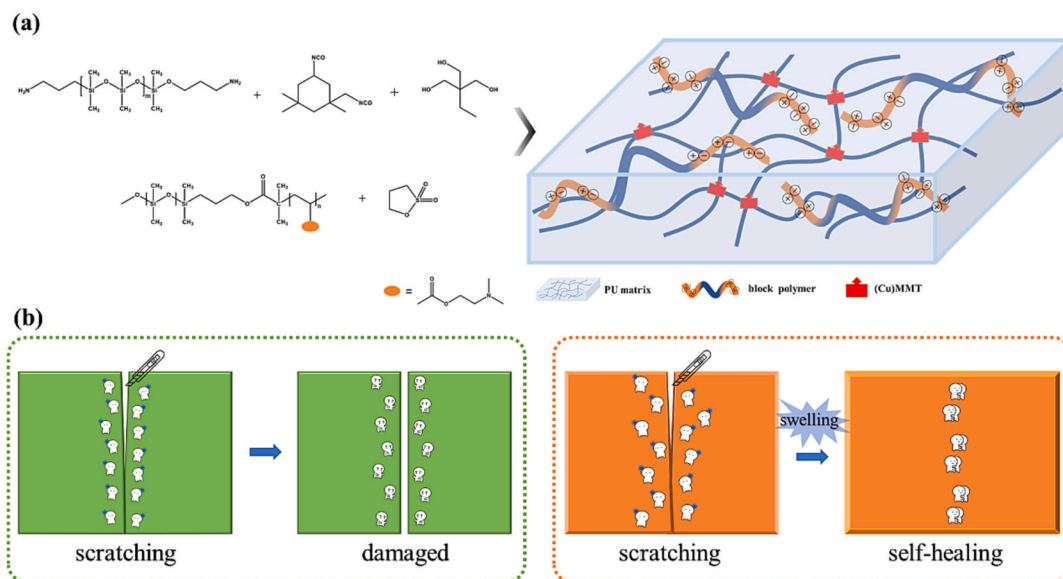


Fig. 1. Preparation and self-healing schematic diagram of antifouling coating.

3.3. Fourier transform infrared spectroscopy (FTIR)

The FTIR spectrum was obtained on an FTIR spectrometer (Nicolet 8700, Thermo Fisher, USA) using the KBr disk method and attenuated total reflection. The scanning range was $650\text{--}4000\text{ cm}^{-1}$ and the spectral resolution was 4 cm^{-1} .

3.4. X-ray photoelectron spectroscopy (XPS)

XPS was performed on a Thermal Scientific K-Alpha spectrometer equipped with a monochromated Al K X-ray source (Al K α , 1486.6 eV). Survey and high-resolution spectra were obtained at the takeoff angles ($\theta = 90^\circ$) relative to the sample surface.

3.5. Atomic force microscope

The surface morphology of coatings was analyzed by Pio Image software and atomic force microscopy (Agilent 5500, BioScope Catalyst, USA) in tapping mode with a silicon tip ($160\text{ }\mu\text{m}$, 165–175 kHz) and a nominal spring constant of 40 N/m.

3.6. Water contact angle (WCA)

WCA measurement was conducted at room temperature on a telescopic goniometer (OCA40, Dataphysics, Germany). The volume of the test water drop was $3\text{ }\mu\text{L}$. The data presented were the average of five measurements at different location of each sample.

3.7. X-ray diffraction (XRD)

MMT powder was tested on an X-ray single-crystal diffractometer (APEX II DUO, Bruker), and the interlayer spacing of the MMT crystal plane after organic modification was calculated according to Bragg's Law:

$$2d \sin \theta = n\lambda$$

where d is the interplanar spacing, θ is the angle between the incident X-rays and the corresponding crystal plane, and n is the reflection order.

3.8. Thermogravimetric analysis (TGA)

TGA (NETZSCH TG209F1, Germany) was conducted under a

nitrogen atmosphere with a heating rate of 10 K/min from room temperature to $800\text{ }^\circ\text{C}$.

3.9. Water swelling test

The swelling degree of different sample was determined according to the equation:

$$S_w = \frac{m_1 - m_0}{m_0} \times 100\%$$

The m_0 and m_1 are the mass of the dry sample and wet sample, respectively.

3.10. Self-healing test

Self-healing tests were evaluated by scratch recovery. The samples were scratched with a blade in air or underwater at room temperature, the change of scratch-width track was recorded by ultra-deep microscope (DMV6, Laica, Germany) and a polarizing microscope (BX51-P, Olympus, Japan).

3.11. Tensile strength and elongation at break

The tensile strength and elongation at break of samples were measured by Universal Testing Machine (KEXIN, WDW3020, China). Rectangular samples ($50\text{ mm} \times 5\text{ mm} \times 1\text{ mm}$) were tested at a tensile speed of 20 mm/min . The value of tensile strength and elongation at break presented were the average of three measurements with an error of less than 5 %.

3.12. Protein adsorption test

The samples were immersed in $50\text{ }\mu\text{g}\cdot\text{mL}^{-1}$ fluorescein isothiocyanate (FITC)-labeled BSA in the dark at room temperature for 24 h, then cleaned by PBS solution and dried with nitrogen. The FITC-labeled BSA adsorption on the samples was observed with fluorescence microscopy (BX-51, Olympus, Japan), which was excited by an argon-ion laser ranging from 488 to 638 nm.

To quantify the protein adsorption density, the sample ($1\text{ cm} \times 1\text{ cm}$) was pre-wetted in phosphate buffer solution (PBS, pH = 7.4) and incubated in BSA solution for 12 h at room temperature. After being rinsed 5 times with PBS solution, the samples were placed in 1 mL 1 wt% aqueous

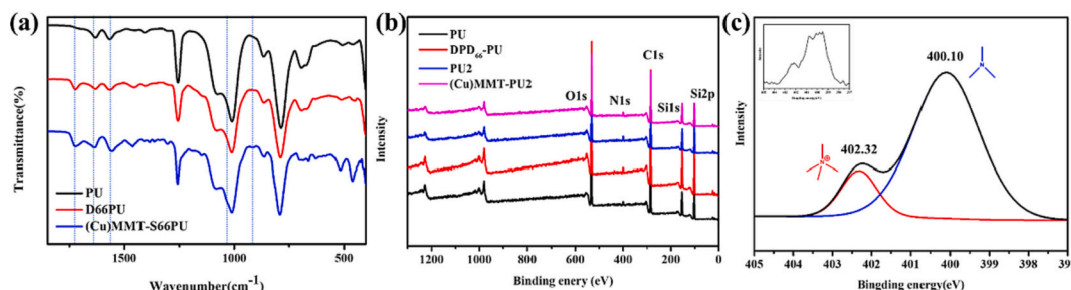


Fig. 2. (a) FTIR spectra of different samples, (b) XPS survey spectra of different samples, (c) N 1s core-level spectra of coating (Cu)MMT-S66PU.

sodium dodecyl sulfate solution shaken for 2 h at room temperature to remove the protein adsorbed on the surface. The elution solution was added to a 96-well plate containing bicinchoninic acid at 37 °C for 30 min. The adsorbed protein density was determined by adsorption intensity at 562 nm using a microplate reader (Multiskan FC, Thermo, America). The resulting data was the mean value of five determinations.

3.13. Biofouling assays

Diatoms were cultured in a climate chamber at 16 °C with a light intensity of 3000 Lux, then 10 mL diatom suspension was added to a petri dish containing glass slides coated with different samples. After incubation at 16 °C for 7 days, the samples were washed slightly with artificial seawater to remove the unattached diatoms. The attachment density of diatoms on the sample surface were visualized by auto-fluorescence image analysis (BX-51, Olympus, Japan) under a halogen lamp and ultra-depth microscope (Leica DVM6, Germany), and the attached diatom density was determined by a hemocytometer. All data were obtained from at least 2 independent experiments with 2 parallel samples, each sample provided 20 measurement points. In addition, the long-term anti-biofouling experiment in the underwater environment was conducted at Jingyue lake (Shanghai, China) for a month.

4. Result and discussion

4.1. Synthesis methodology

A series of amphiphilic block polymer PDMS-PSBMA with different chain length are blended with PDMS-NH₂, isophorone diisocyanate, chain extender and modified montmorillonite, where further zwitterionization of PDMAEMA segments and formation of polyurethane are carried out simultaneously. The specific synthesis process of the amphiphilic polymer and polyurethane matrix is shown in Fig. S3. As shown in Fig. 1, the PU matrix has inherent self-healing properties since the high flexible PDMS segments confer strong migration ability to the polymer, while the polyurea structure allows the formation of high-density reversible hydrogen bonds [27]. PDMS-PSBMA not only endow the resulting self-healing PU coating with anti-biofouling function, but also cause slight swelling underwater through the hydrophilic segment, which increases the reorganization probability of reversible hydrogen bonds and further improves self-healing efficiency of the coatings. Meanwhile, the intercalated MMT and PU are bonded together through coordination bonds between copper ions and unsaturated electron pairs, and the mechanical properties of the composite coating is greatly improved due to the resulting reinforced concrete-like structure.

The Fourier transform infrared (FTIR) spectrum of PU, D66PU, and (Cu)MMT-S66PU is shown in Fig. 2a. The peak at 789 cm⁻¹ belongs to the -Si-O- stretching vibrations [28], which originates from the polydimethylsiloxane segments in the PU backbone. The peaks at 1550–1775 cm⁻¹ are assigned to C=O presented in three samples, and the peaks at 1632 cm⁻¹ and 1567 cm⁻¹ correspond to the urea group in PU, which is formed by the reaction of the amino group in PDMS-NH₂

Table 1
Elemental atomic percentages of different samples from XPS analysis.

Samples	Elemental atomic percent (%)					
	C	O	Si	N	S	Cu
PU	50.44	24.51	21.97	3.08	–	–
D66PU	51.70	22.77	23.14	2.39	–	–
S66PU	50.44	22.66	22.59	2.60	1.71	–
(Cu)MMT-S66PU	51.18	22.28	21.81	2.76	1.77	0.19

with isocyanate group. The new peak at 1723 cm⁻¹ in D66PU spectra is attributed to the ester group in PDMS-PDMAEMA, which demonstrates the successful incorporation of the amphiphilic polymer into the PU matrix. The peaks appearing at 923 cm⁻¹ and 1036 cm⁻¹ in spectra of (Cu)MMT-S66PU are assigned to quaternary amine groups and sulfonyl groups, which indicates the successful quaternization of the tertiary amine in the PDMAEMA segment to form the zwitterionic PSBMA structure. Peaks at 518 cm⁻¹ and 461 cm⁻¹ are the splitting peaks of Si–O bending vibration in MMT [29]. Thus, the FTIR spectrum proves the successful introduction of PDMS-PSBMA and MMT in PU.

As shown in Fig. 2b and Table 1, four major emission peaks at 284.8, 532.1, 102.3, and 400.1 eV are ascribed to C 1s, O 1s, Si 2p, and N 1s, respectively. A little sulfur is detected in S66PU and (Cu)MMT-S66PU, i. e. 1.71 % and 1.79 %, respectively, which is attributed to the reaction of the quaternary amine group with 1,3-propane sultone. The nitrogen content detected in PU is 3.08 %, which is dropped to 2.39 % in D66PU and gradually increased with increasing molecular weight of PDMS-PDMAEMA (Fig. S4 and Table S2). As shown in Fig. 2c, the peak at 400.1 eV is assigned to N 1s of the amine group in PDMAEMA segments and PU, and the peak at 402.2 eV is the characteristic peak of ionized nitrogen atom (N⁺). These results suggest that the tertiary amine in the PDMAEMA segment has been successfully transformed to positively charged quaternary amine in PSBMA via the treatment with 1,3-propane sultone [30].

Atomic force microscope is used to explore the 3D surface topography of the coatings before and after being immersed in water. As shown in Fig. 3a–d, all coatings are macroscopically homogeneous, since the amphiphilic polymer and PU can integrate sufficiently due to their identical PDMS segment. The surface topography of D66PU does not exhibit a pronounced change before (Fig. 3a) and after (Fig. 3b) immersion in water, whose root mean square roughness (Sq) value is increased just from 0.359 nm to 0.371 nm, indicating that PDMAEMA segment has a slight effect on the surface structure. As compared with D66PU, the roughness of S66PU has shown a significant increasing trend after immersion (Fig. 3c, d), where the Sq value is more than 4 times larger from 0.563 to 2.68 nm. The high hydrophilicity of PSBMA segments results in a faster migration rate to the surface underwater and a higher roughness is generated [31]. Furthermore, the surface topography images reveal an increase in roughness with increasing PSBMA content, as S165PU shows high degree of phase separation (Fig. S5). Interestingly, excessive roughness is more likely to induce biofouling formation as proved by subsequent antifouling tests.

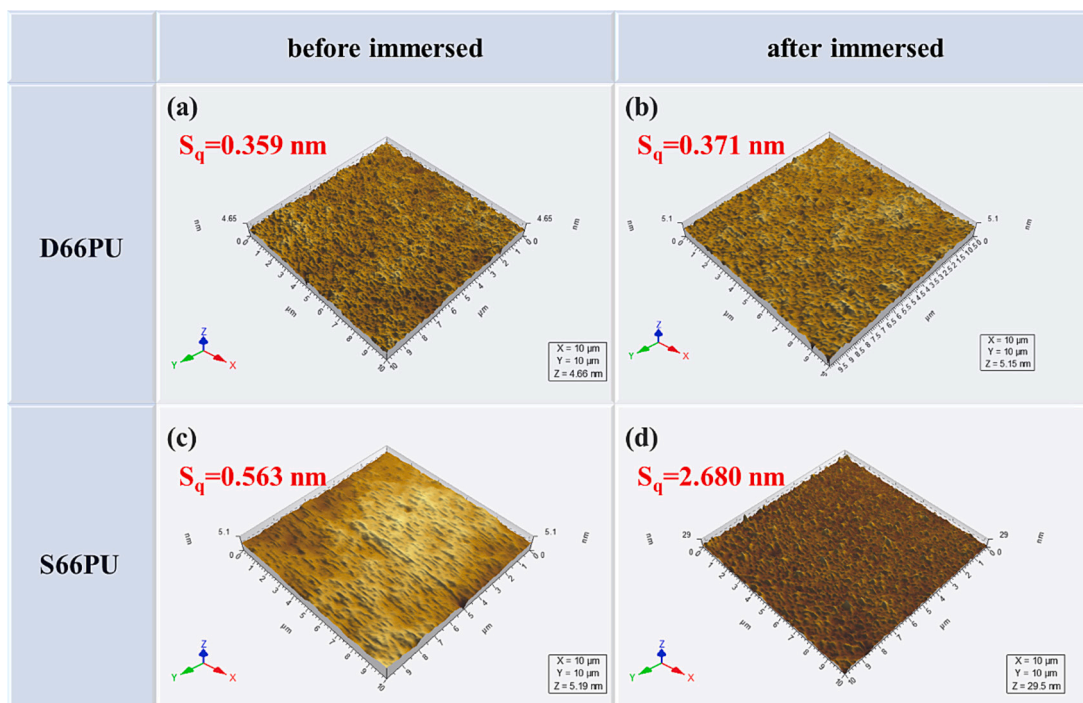


Fig. 3. 3D surface topography of D66PU (a, b) and S66PU (c, d) coatings before and after immersed in water for 24 h by AFM observation.

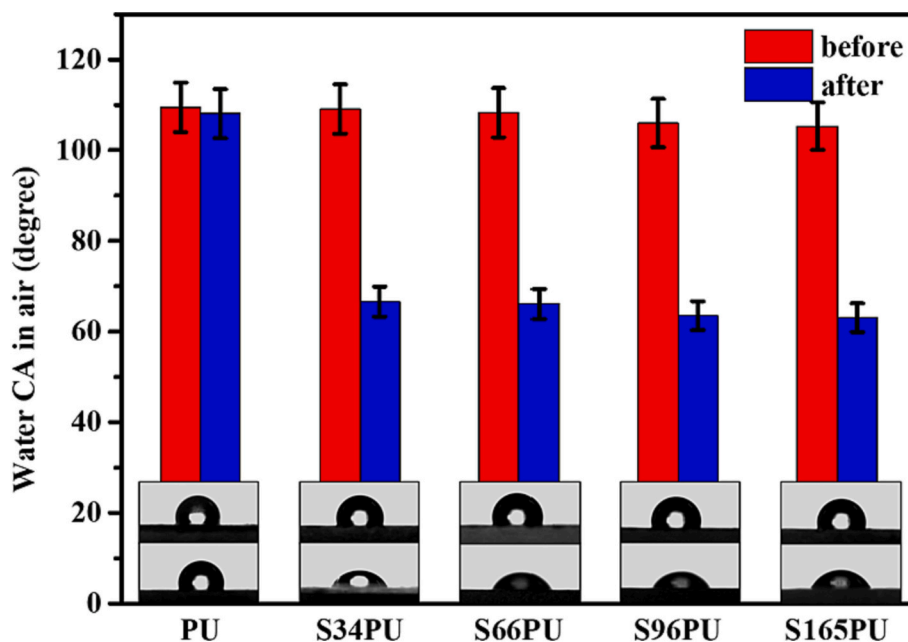


Fig. 4. WCA on different coatings before and after immersed in water for 24 h.

As shown in Fig. 4, all coatings present hydrophobic state in the air due to the low surface energy of PDMS segments. After 24 h of immersion in water, the water contact angle (WCA) of PU has a minor change from 110° to 108°, while the WCAs of other coatings exhibit a significant decrease, i.e. from 109° to 67° of S34PU, 108° to 62° of S66PU, 106° to 64° of S96PU and 105° to 61° of S165PU, respectively. This is because the hydrophilic PSBMA segments gradually migrate to the coating surface during immersion underwater. The WCA result is consistent with AFM, which jointly proves the effect of hydrophilic segments length on surface topography and wettability.

4.2. Mechanical properties

As shown in Fig. 5, the tensile strength of SxPU is decreased from 0.90 MPa to 0.68 MPa and the elongation at break is increased from 348 % to 466 % with the increase of the PDMS-PSBMA molecular weight. Correspondingly, the glass transition temperature (T_g) of the coating is gradually decreased from -10.9 °C to -14.4 °C. PDMS-PSBMA can be considered as a plasticizer to weaken the force between PU molecular chains, leading to an increase of sample free volume, which expands with the increase of the polymer molecular weight. In addition, the only one T_g in S34PU to S96PU shows excellent compatibility between PU

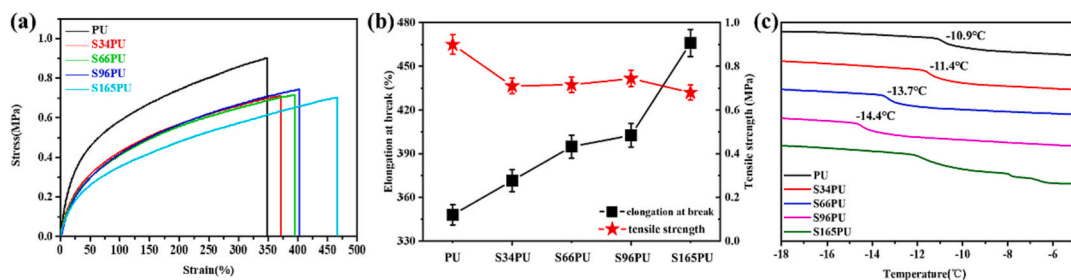


Fig. 5. (a) Stress-strain curves, (b) tensile strength and elongation at break, (c) T_g of PU, S34PU, S66PU, S96PU and S165PU.

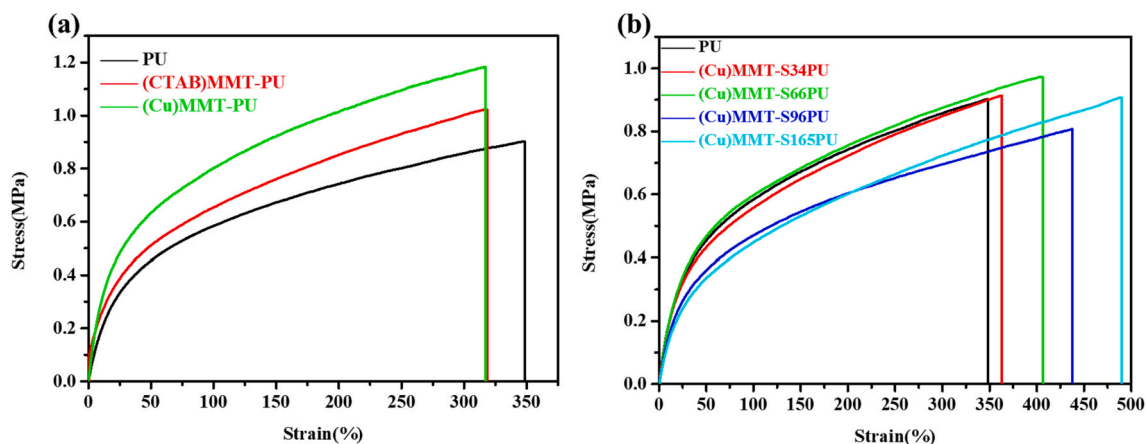


Fig. 6. (a) Stress-strain curves of PU, (CTAB)MMT-PU, and (Cu)MMT-PU; (b) stress-strain curves of PU, (Cu)MMT-S34PU, (Cu)MMT-S66PU, (Cu)MMT-S96PU, and (Cu)MMT-S165PU.

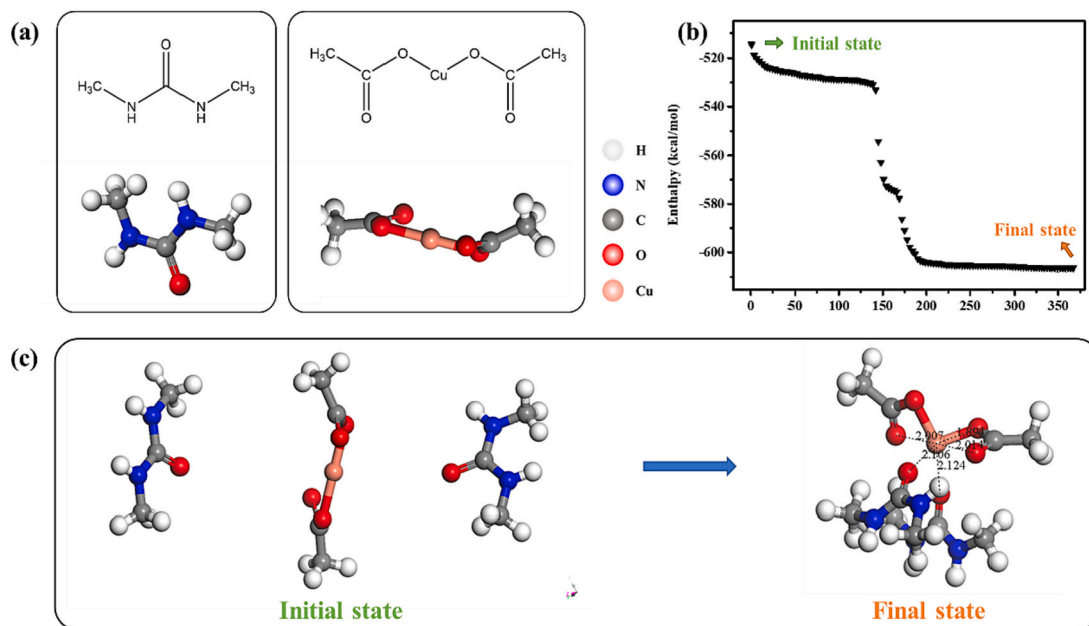


Fig. 7. (a) The molecular formula structure of N,N' -dimethylurea and copper acetate, (b) the energy calculation process, (c) the initial and final state of polyurethane and copper acetate (taking two polyurethane fragments and a copper acetate molecule as an example for simulation).

matrix and PDMS-PSBMA, while the spectrum of S165PU is a multi-step curve since long PSBMA segment leads to pronounced incompatibility.

In order to compensate for the mechanical property loss caused by polymer addition, montmorillonite (MMT) is incorporated into PU to achieve enhanced function. As it is difficult for inorganic MMT to integrate well with polyurethane, modification of MMT is carried out and

copper ions are introduced subsequently (Fig. S6 and Table S3). As shown in Fig. 6a, the tensile strength of (CTAB)MMT-PU and (Cu)MMT-PU is 1.02 MPa and 1.18 MPa respectively, which is 13 % and 31 % higher than that of PU. According to Fig. 6b, (Cu)MMT-S66PU has the best comprehensive mechanical property with tensile strength of 0.97 MPa and breaking elongation of 406 %.

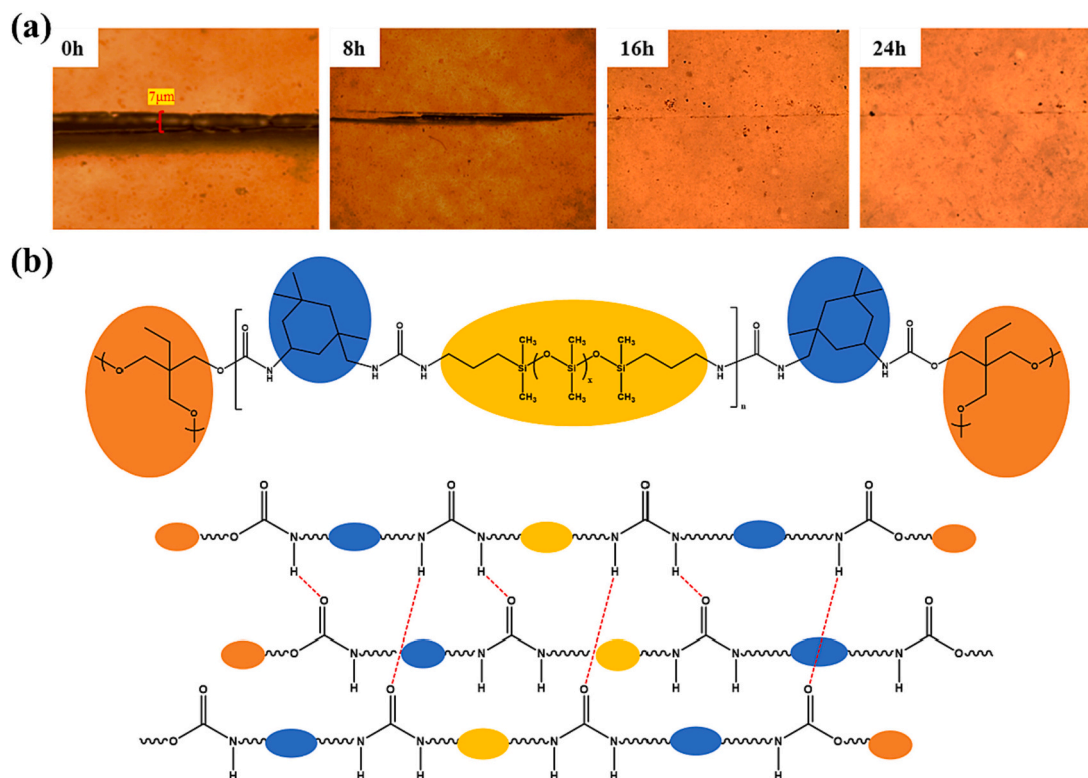


Fig. 8. (a) Self-healing process of S66PU at room temperature imaged by polarizing microscope for 24 h, (b) the possible forms of reversible hydrogen bonding in coatings.

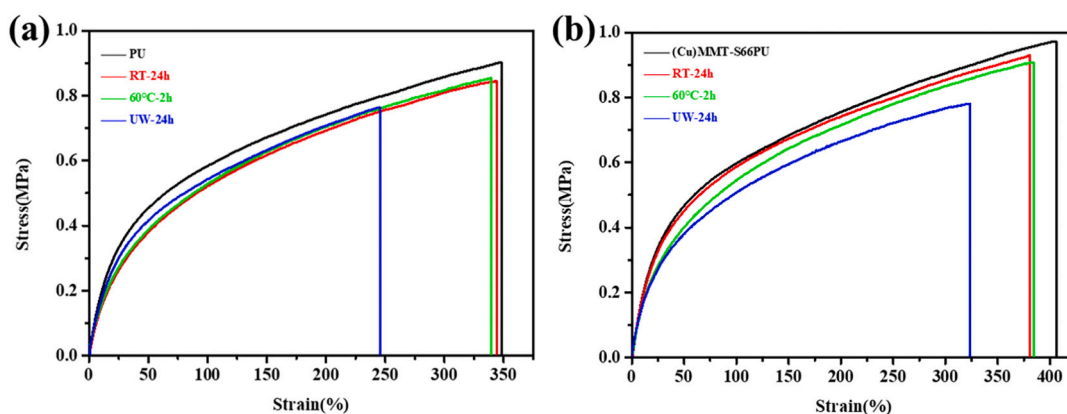


Fig. 9. Stress-strain curves of (a) PU and (b) (Cu)MMT-S66PU after self-healing at 25 °C, 60 °C and underwater, respectively.

The introduction of (Cu)MMT has greatly improved the mechanical property of PU, and molecular simulation is carried out to reveal the enhancement mechanism. Fig. 7a shows the geometrically optimized model, where copper particles in MMT are represented by copper acetate, while the polyurethane segment is represented by *N,N'*-dimethylurea with the same -NH-CO-NH- structure. The energy change in the dynamic simulation process is shown in Fig. 7b. The original energy of the system is -515 kcal/mol, and the lowest energy state (that is, the most stable state) with an energy of -607 kcal/mol is determined among a large number of possible conformations through simulations. The molecule structure at the lowest energy state in Fig. 7c shows that coordination bonds are formed between copper ions and the oxygen atoms in the polyurethane segment [32], resulting in the mechanical property strengthening.

4.3. Self-healing of coatings

It is clearly observed that the crack disappears as well as the sample restores to its original state completely at room temperature after 24 h (Fig. 8a). The original scratch depth is about 50 μm as same as the coating thickness, which has been decreased to about 1 μ after being repaired at room temperature for 16 h (Fig. S7), and the coating surface is completely uniform after 24 h self-healing process. The relevant self-healing mechanism is illustrated in Fig. 8b, a large number of reversible hydrogen bonds are formed between ureido ester groups, which are broken when encountering mechanical damage, and then recombined through the motion of macromolecules to achieve self-healing effect.

The self-healing effect of samples under different conditions is evaluated by stress-strain tensile test. As shown in Fig. 9a, PU and Cu (MMT)-S66PU can almost restore to its initial tensile strength after 24 h of self-healing at room temperature. Despite the improved mechanical

Table 2

The final self-healing efficiency^a of PU and (Cu)MMT-S66PU under different conditions.

	RT-24 h	60 °C-2 h	UW-24 h
PU	90.2 %	89.9 %	58.3 %
(Cu)MMT-S66PU	89.5 %	86.8 %	63.5 %

^a The repair efficiency is calculated by the integral area of the stress-strain curve before and after self-healing.

Table 3

Swelling degree of different coatings in water.

Samples	Swelling degree ^a (wt%)
PU	0.06
S34PU	3.63
S66PU	4.23
S96PU	4.24
S165PU	5.91

^a Calculated by eq $W_{\text{swelling}}\% = (m_2 - m_1) / m_1 * 100\%$.

strength of (Cu)MMT-S66PU, its self-healing efficiency is slightly lower than that of PU. This phenomenon may be due to the stronger effect of (Cu)MMT on PU compared with PDMS-PSBMA on PU. A certain cross-linking structure is formed between Cu(MMT) and PU matrix through the formation of coordination bonds, leading to an increase in mechanical properties of the coating. On the other hand, the cross-linking structure limits the mobility of macromolecular segment, which further affects the recombination of reversible hydrogen bonds and leads to a decrease in self-healing efficiency of coating. As shown in Fig. 9b, it takes both PU and Cu(MMT)-S66PU only 2 h to restore its original state at 60 °C, since the macromolecular segments move faster with increasing temperature, so as to increase the efficiency of reversible bonds recombination.

Typically, the self-healing effect achieved by reversible bonds is

greatly reduced underwater, since such bonds prefer to combine with water molecules instead of themselves [33], leading to self-healing efficiency reduction. As a result, the original mechanical strength of PU cannot be completely restored underwater. Interestingly, the self-healing efficiency of Cu(MMT)-S66PU is significantly higher than that of PU as shown in Table 2. Due to the high water-binding ability of PSBMA segments, the minor swelling degree of the sample (Table 3) and contact probability of the macromolecular reversible bonds are increased. Therefore, the underwater self-healing efficiency is improved.

4.4. Antifouling tests

Marine biological pollution formation originates from the attachment of nutrients such as proteins to submerged substrate. As shown in Fig. 10a, the PU coating surface shows a large area of bright green, while SxPU surface shows almost no green, especially S66PU appears completely black, indicating a low degree of FITC-labeled BSA protein adsorption. Furthermore, the protein adsorption on the coating surface is measured quantitatively by a microplate reader assay. As shown in Fig. 10c, the adsorption density of BSA protein on SxPU is dropped sharply from 156 $\mu\text{g}/\text{cm}^2$ for PU to less than 30 $\mu\text{g}/\text{cm}^2$. The least adhesion density of 14 $\mu\text{g}/\text{cm}^2$ is observed for S66PU, indicating its excellent resistance to nonspecific protein adsorption. The reason is that the zwitterionic PSBMA segments in coatings gradually migrate to the surface in the underwater environment, thereby a dense hydration layer formed around the coating surface effectively resists protein adsorption. In addition, the microphase separation structure generated by the segment migration forms a nano-scale complex surface morphology on the coating surface as proved by previous AFM characterization, so as to resist the adhesion of non-specific proteins.

As shown in Fig. 10b, the amount of adsorbed diatom on SxPU is significantly decreased as compared with PU coating. The minimum adhesion density of S66PU is $18 \times 10^3 \text{ n}/\text{cm}^2$, which is significantly

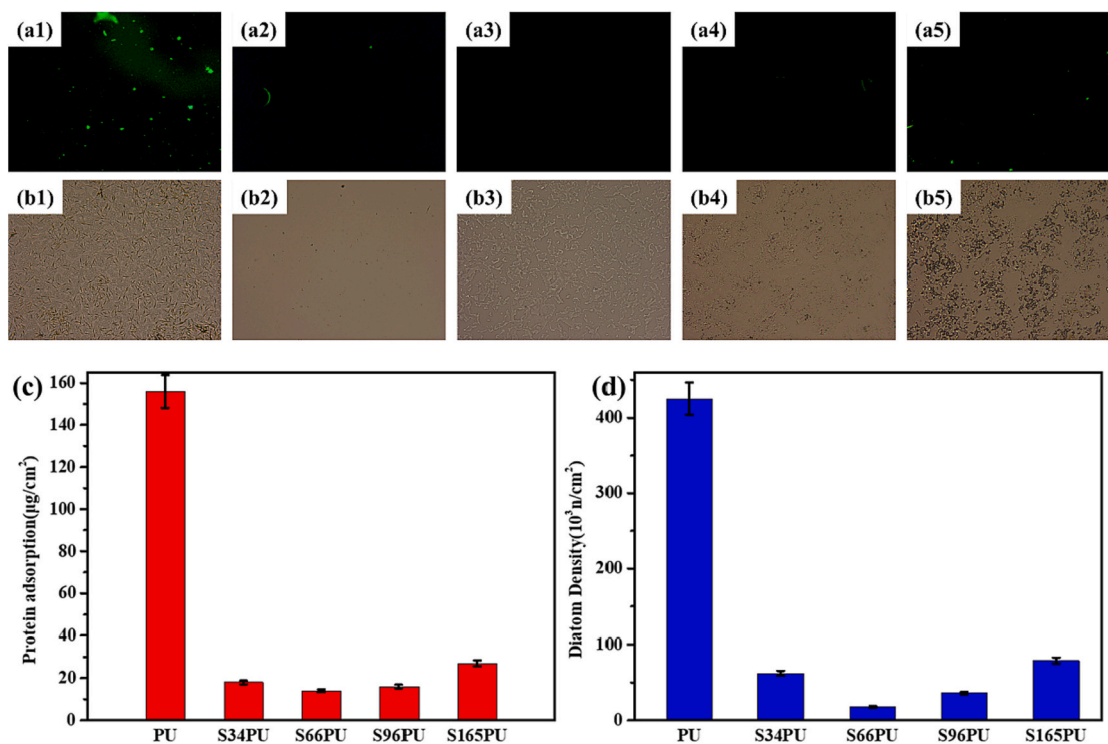


Fig. 10. Fluorescence microscopy images of BSA adsorption (a) and microscope images of the attachment of *P. tricornutum* (b) on PU, S34PU, S66PU, S96PU, and S165PU; (c) BSA protein adsorption density on coatings, (d) density of *P. tricornutum* attached on coating surfaces after 7 days. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

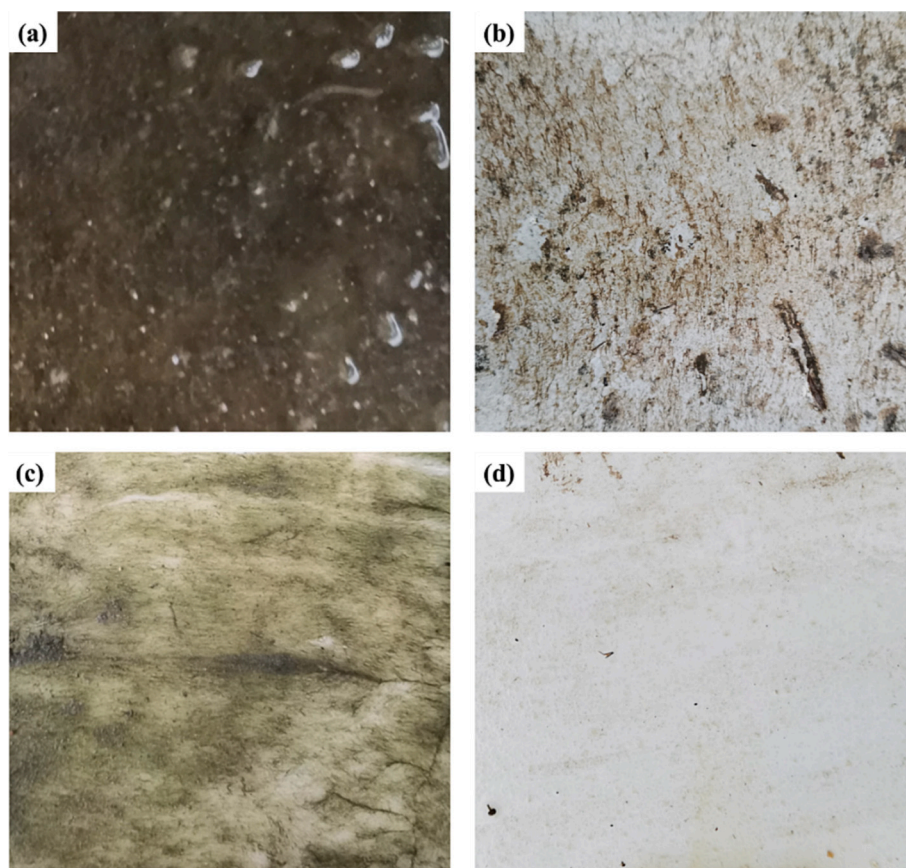


Fig. 11. Images of panels after immersion underwater for a month. (a) Blank substrate without coating as a control, (b) panel coated with the commercial sylgard 184 coating (c) panel coated with PU, (d) panel coated with (Cu)MMT-S66PU.

lower than $425 \times 10^3 \text{ n/cm}^2$ of PU control (Fig. 10d). The unique heterogeneous rough morphology at a micro-nano level on the coating surface impedes the sedimentation of diatoms. Furthermore, the dense hydration layer increases the difficulty of fouling organisms to attach, since much more energy is required to penetrate through the hydration layer. Interestingly, the adhesion density of diatom is not gradually decreased with molecular weight increase of the added PDMS-PSBMA, since the adhesion density of S96PU and S165PU shows a slight upward trend, which is $36 \times 10^3 \text{ n/cm}^2$ and $79 \times 10^3 \text{ n/cm}^2$, respectively. This result clearly shows that the fouling organism is more likely to attach on the coating surface with larger roughness [34]. And the resulting antifouling coating greatly inhibits the formation of early biofouling underwater.

To verify the long-term effectiveness of the composite coating in real environment, the field test is used to evaluate its antifouling performance for a month. As shown in Fig. 11a, the surface of the control epoxy panel is completely covered by fouling organisms such as algae and bryozoans, etc., indicating heavy fouling. The commercial sylgard 184 coating in Fig. 11b indicates an improved antifouling effect. Some organisms located PU surface as shown in Fig. 11c are significantly less than those on the blank control. Little fouling is observed on the surface of coating (Cu)MMT-S66PU, indicating excellent antifouling performance. This is attributed to the amphiphilic microphase-separated surface, which endows coating anti-adhesion performance and significantly reduces the attached fouling organisms.

5. Conclusion

In this study, a composite multifunctional coating with high mechanical strength, excellent self-healing performance, and anti-biofouling function is prepared successfully, via the introduction of

amphiphilic polymer SPSx and modified montmorillonite into a self-healing polyurethane network. The addition of modified montmorillonite significantly enhances the coating's mechanical strength under the premise of ensuring antifouling performance, and the self-healing efficiency underwater is significantly improved through coatings micro swelling. The coating has a remarkable antifouling effect both in the initial stage of fouling formation and actual complex environment. It is believed that the resulting research will be a pioneering effort and lay the foundation for future large-scale application of antifouling coatings, due to its extremely simplified experimental process and outstanding comprehensive performance.

CRediT authorship contribution statement

Ziyuan Liu: Conceptualization, Investigation, Data curation, Writing – original draft, Visualization. **Chunju He:** Methodology, Writing – review & editing, Funding acquisition, Supervision.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

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